KANDLE: Grid-Adaptive Kolmogorov-Arnold Networks

Ilse Vermeer, Rohan Kapoor, Yuki Sato (Meridian Mathematics Group)

Abstract

Kolmogorov-Arnold networks (KANs) replace the fixed activation functions of a multilayer perceptron with learnable univariate functions. We present KANDLE, a KAN whose B-spline activations refine their grids during training. On the Feynman equations benchmark KANDLE reaches a median RMSE of 1.2e-4 with ten times fewer parameters than an MLP of similar accuracy.

1 Introduction

KANs are based on the Kolmogorov-Arnold representation theorem, which states that every continuous multivariate function on a bounded domain can be written as a finite composition of continuous functions of one variable and addition. A KAN layer generalises this construction to arbitrary widths and depths.

In a multilayer perceptron the nodes apply fixed nonlinearities and the edges carry scalar weights. In a KAN the learnable activation functions sit on edges and the nodes only sum their inputs, so every edge is a small function that can be plotted and, often, recognised.

2 Method

Each edge function is a cubic B-spline plus a residual SiLU term. The spline has order k = 3 and G grid intervals over the range of its input:

$$\phi(x) = w_b \, \mathrm{silu}(x) + w_s \sum_i c_i B_i(x) \qquad (1)$$

Coarse grids train quickly but limit accuracy, and fine grids from the start train slowly. KANDLE uses grid extension from G = 5 to G = 20 in three stages: when the loss plateaus, each spline is refitted by least squares on a finer grid, so training continues from the coarse solution instead of restarting.

After training, edges whose L1 norm falls below a threshold are pruned, and the remaining edge functions can be matched against a library of symbolic forms such as sin, exp and x^2.

3 Results

Table 1: Median test RMSE on the Feynman equations benchmark.

Model	Parameters	RMSE
MLP (4 layers)	12,400	3.1e-4
KAN (fixed grid G = 5)	1,180	9.6e-4
KANDLE (G = 5 to 20)	1,240	1.2e-4

Grid extension accounts for most of the improvement over a fixed-grid KAN. Training is slower per step than for an MLP, by a factor of about 6, because each edge evaluates a spline.

References

[1] V. Arnoldsen. On the representation of continuous functions of several variables. Mathematical Notes, 1957.

[2] C. de Boor-Lind. A practical guide to splines for learning. Numerical Methods Press, 2001.